

OPERATING SYSTEMS

UE25CS242A

Scheduling Algorithms

Dr. Anitha K
Department of Computer Science & Engineering

Acknowledgement, Prof. Pavan A C

Slides Credits for all the PPTs of this course

- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources:



1. "Operating System Concepts", Abraham Silberschatz, Peter Baer Galvin, Greg Gagne 10th Edition, John Wiley & Sons, Global Edition, 2023.
2. "Advanced Programming in the Unix Environment", Richard Stevens and Stephen A Rago, Pearson, 3rd edition, 2017.
3. Some conceptual text and diagram from "Operating Systems, Internals and Design Principles", William Stallings, 9th Edition, Pearson, 2018

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Multi-level Queue & Feedback Queue Scheduling

Dr. Anitha K
Department of Computer Science and Engineering

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Multilevel Queue

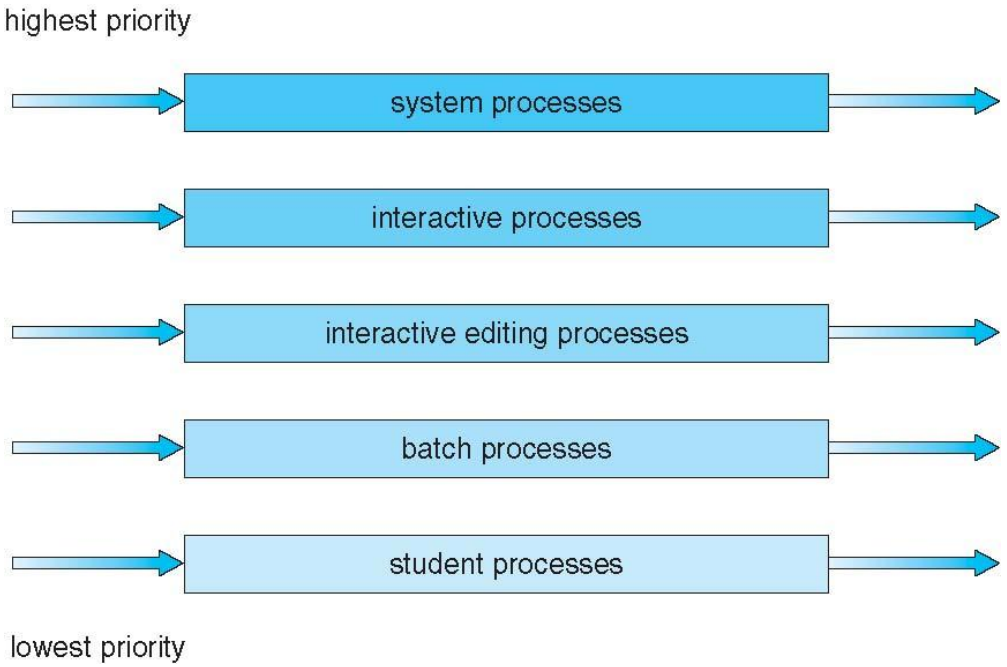


- Ready queue is partitioned into separate queues:
  - **foreground** (interactive)
  - **background** (batch)
- Process permanently assigned to a queue
- Each queue has its own scheduling algorithm:
  - foreground – RR
  - background – FCFS
- In addition, scheduling must be done between the queues
  - Fixed priority scheduling; (serve all from foreground then from background). Possibility of starvation.
  - Time slice – each queue gets a certain amount of CPU time which it can schedule

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Multilevel Queue Scheduling

- Each queue has absolute priority over lower-priority queues
- No process in the batch queue, for example, could run unless the queues for system processes, interactive processes, and interactive editing processes were all empty
- If an interactive editing process entered the ready queue while a batch process was running, the batch process would



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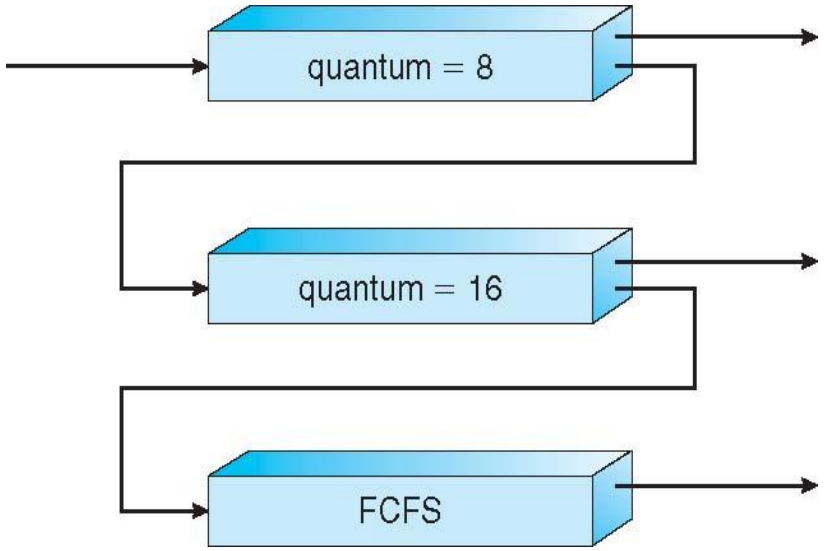
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Multilevel Feedback Queue



- A process can move between the various queues; aging can be implemented this way
- Multilevel-feedback-queue scheduler defined by the following parameters:
  - Number of queues
  - Scheduling algorithms for each queue
  - Method used to determine when to upgrade a process
  - Method used to determine when to demote a process
  - Method used to determine which queue a process will enter when that process needs service

Example of Multilevel Feedback Queue



Multilevel feedback queues

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Multilevel Feedback Queue



- Three queues:
  - Q_0 – RR with time quantum 8 milliseconds
  - Q_1 – RR time quantum 16 milliseconds
  - Q_2 – FCFS

  - The scheduler first executes all processes in queue 0.
  - Only when queue 0 is empty will it execute processes in queue 1.
  - Processes in queue 2 will be executed only if queues 0 and 1 are empty.
  - A process that arrives for queue 1 will preempt a process in queue 2.
  - A process in queue 1 will in turn be preempted by a process arriving in queue 0.



THANK YOU

Dr. Anitha K
Department of Computer Science and Engineering
anithak@pes.edu